

Corporate & Institutional Banking

Group exposure is the hard problem — and it is a data-model problem, not a reporting one

Every corporate bank can report exposure by counterparty. Far fewer can answer what the group actually owes when the parent, its subsidiaries, its guarantors and its cross-holdings sit in different systems, segments and countries.

1

question that is genuinely hard

5

entity relationships that must exist

4

ways the answer silently goes wrong

12

countries in the reference deployment

\$40B+

reference asset size

0

reporting layers that fix a bad model

DRAWN FROM CORPORATE BANKING DELIVERY · GULF AND REGIONAL INSTITUTIONS

Group hierarchy and related parties · credit committee workflow · multi-country consolidation · exposure roll-up and limits

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Group exposure fails in the data model and gets discovered in the reporting layer — which is why it is usually addressed in exactly the wrong place

Situation

A corporate bank needs to answer a simple-sounding question: what is our total exposure to this group? It underpins limit setting, concentration appetite, credit committee decisions and regulatory reporting.

Every bank has a number. The question is whether it is right.

Complication

The number is assembled from facilities booked against legal entities. Getting from those to a group requires knowing which entities belong to which group, who guarantees whom, which cross-holdings create indirect exposure, and how all of that changes over time.

None of that lives in a lending system. It lives in a relationship model that most architectures treat as reference data, maintained manually, and never versioned.

The position

Group exposure is computable only if the entity relationships are first-class, owned, versioned and complete. That is a data-model commitment made early, not a consolidation query written later.

A reporting layer over an incomplete relationship model does not produce a wrong number that looks wrong. It produces a confident number that is quietly incomplete — which is worse.

THE NUMBERS THAT MATTER

5

relationship types required

4

silent failure modes

1

owner the model needs

0

queries that fix bad data

100%

of the model is relationship data

1

point-in-time requirement

Why this is the corporate banking problem: in retail, exposure aggregates cleanly because a customer is a person and a person is one entity. In corporate, the thing the bank is exposed to is a structure — and structures are not stored by lending systems, which book facilities to counterparties.

This is a domain analysis drawn from my own corporate banking work, with a comparable public deployment used only for scale context

PUBLICLY REPORTED Reference context Scale figures from a comparable Gulf corporate banking deployment, used to establish the size of the problem. Not a client of mine. \$40B+ asset size 12 countries deployed 300+ platform users 10+ systems integrated - Corporate and group 360 scope - Credit committee workflow in scope	DELIVERY RECORD My ownership Corporate and wholesale banking programmes delivered at Gulf and regional institutions between 2018 and 2023. - Group and related-party model design - Wholesale deal and credit approval - Corporate Client 360 across six countries - Exposure roll-up and limit structures - Credit committee workflow delivery - Related-party data quality assessment	ILLUSTRATIVE Modelled Structural examples and failure-mode frequencies illustrate the argument. No real group or exposure appears. - Group structure worked example - Failure-mode decomposition - Data completeness illustration - Replace with portfolio actuals - Shape is the argument - Method transfers unchanged	OUT OF SCOPE Not claimed Explicitly excluded. In particular, no delivery claim is made on the reference deployment. - Delivery of the reference deployment - Any client, group or exposure data - Any bank's limit or appetite framework - Any credit committee decision - Any confidential document - Measured business outcomes
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THE QUESTION

Five relationship types have to exist before the question is answerable — and most architectures model two of them properly

Ownership

Parent to subsidiary, with percentage and effective control. Determines what consolidates into the group at all.

Usually modelled

Guarantee

Who stands behind whose obligation. Creates exposure to an entity that has borrowed nothing.

Usually modelled

Cross-holding

Entities holding stakes in each other, or in a common third party. Creates indirect exposure that no direct query finds.

Rarely modelled

Common control

Entities with no ownership link but the same controlling party. Regulators frequently treat these as one exposure.

Rarely modelled

Economic dependence

Entities whose viability depends on another — a sole supplier, a single-tenant property, a captive distributor.

Almost never modelled

Why the last three are the ones that matter

Ownership and guarantee are modelled almost everywhere, because they are visible in the credit file and required for booking. They also produce the exposure number that everyone already has.

Cross-holding, common control and economic dependence are the ones that make a group exposure surprising — and a surprising group exposure is the entire point of measuring it. A bank that models only the first two has built a system that confirms what it already believed, at some cost, and will still be surprised in a downturn by exactly the concentrations it was trying to detect.

HOW IT GOES WRONG

Four failure modes, and none of them produces an error — each produces a confident number that is quietly incomplete

01 Incomplete relationship data

The model supports cross-holdings; nobody populated them. The query runs, returns a number, and omits an entire limb of the group.

THE TELL

The group exposure equals the sum of the obvious subsidiaries — exactly what a manual analyst would have produced.

02 Point-in-time collapse

The hierarchy is stored as current state with no history. Last quarter's committee decision cannot be reproduced because the structure has since changed.

THE TELL

Historical reports cannot be regenerated and nobody can explain why the number moved.

03 Two systems, two hierarchies

The lending system and the relationship platform each resolve groups their own way. Both are internally consistent and they disagree.

THE TELL

Reconciliation differences that shift each month and are attributed to timing.

04 Double counting through guarantees

A guaranteed facility is counted at the borrower and again at the guarantor, inflating group exposure — or netted incorrectly and understating it.

THE TELL

Group exposure exceeds the sum of facilities, or falls below the largest single facility.

None of the four throws an exception. All four are found by someone who already suspected the number was wrong — which is why the tells matter more than the validations.

Six properties make group exposure computable — and five of them are decided in the data model, not in the query

Property	What it means in practice	Decided in	Retrofit?
Relationships are first-class	Ownership, guarantee, cross-holding and control are objects with their own attributes and lifecycle — not fields on an entity	Data model	Hard
The hierarchy is versioned	Every structure change is a dated event; any past view can be reconstructed exactly	Data model	Very hard
One resolution authority	Exactly one system decides which entities form a group; all others consume that answer	Architecture	Hard
Exposure is typed	Direct, contingent, guaranteed and indirect are distinguished, so netting rules can be applied correctly	Data model	Hard
Completeness is measured	The share of entities with verified relationship data is reported alongside the exposure number itself	Governance	Easy
An owner exists	A named function is accountable for relationship data quality, with a review cycle	Governance	Easy

The two that are easy and skipped

- Completeness measurement and named ownership cost almost nothing and are routinely absent.
- Reporting completeness alongside the number is the single cheapest credibility intervention available.
- Ninety-four per cent verified is far more useful than a bare figure.

The one that is genuinely very hard

- Versioning the hierarchy is difficult to retrofit because the history does not exist to backfill.
- The cost is invisible until a past committee decision must be reproduced.
- It is worth doing on day one purely because it cannot be done on day one thousand.

How to sequence the six

- Do both governance properties immediately — they are cheap and they reveal how bad the data is.
- Do versioning and typing in the initial model, because neither is recoverable later.
- Resolution authority is a decision, not a build — make it early.

The credit committee is where an incomplete group model becomes a decision — which is why the exposure view has to carry its own confidence

01

Proposal assembled

The facility, the counterparty and the stated group exposure are brought together for the committee pack.

02

Group position shown

Total exposure across the group, broken by type — direct, contingent, guaranteed, indirect — with completeness stated.

03

Concentration tested

Group position against limit, against appetite, and against related-sector concentration.

04

Conditions set

Approval with conditions, and those conditions attached to the group rather than only to the facility.

05

Position updated

The decision changes the group's committed exposure immediately, not at the next reporting cycle.

Why completeness belongs on the pack

- A committee taking a limit decision needs to know how much of the group it can actually see.
- Showing exposure without completeness invites the number to be treated as exact when it is an estimate with unknown error.
- Stating it is not an admission of weakness — it is the difference between a measurement and an assertion.

Conditions attach to the group

- A condition set on a facility is monitored on that facility. A condition about group leverage or group concentration has to attach to the group.
- If the model cannot hold a group-level condition, it will be recorded as free text and never monitored.
- This is the most common gap between what a committee decided and what the system tracks.

The update must be immediate

- Committed exposure changes when the committee decides, not when the facility books.
- A bank whose group position updates on booking is blind during the gap, which is exactly when a second proposal on the same group arrives.
- Modelling committed separately from drawn is what closes it.

Four judgements about group exposure that carry to any corporate bank

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| 01 | Treat relationships as objects, not attributes | Ownership, guarantee, cross-holding and common control need their own lifecycle, evidence and versioning. Modelled as fields on an entity they cannot be dated, sourced or verified — and a group view built on them cannot be defended. |
| 02 | Version the hierarchy on day one | It is the only property on the list that genuinely cannot be retrofitted, because the history does not exist to backfill. The cost of skipping it is invisible until a past committee decision has to be reproduced. |
| 03 | Publish completeness with the number | A group exposure shown as a bare figure invites treatment as exact. Shown with the share of the group verified, it becomes a measurement with a known limitation — which is what a committee can actually use. |
| 04 | Give the group somewhere to hold a decision | Conditions, limits and appetite are set at group level. If the model cannot carry them there, they become free text in a minute and are never monitored — the most common gap between what was decided and what is tracked. |

Every corporate bank can report exposure by counterparty. The ones that can answer for the group have made a data-model commitment years earlier — and no reporting layer, however good, substitutes for it.